Qurious

Learn quantum computing by building it

SMART INDIA HACKATHON 2026 · SIH26140

How Qurious Compares

An honest comparison against every significant quantum-learning product surveyed — what they do well, where they stop, and the eight things Qurious does that none of them do.

Field	Detail
Problem statement	SIH26140 — AI-Based Interactive Quantum Algorithm Learning Platform
Organisation	Egreen Quanta · Theme: Smart Education · Category: Software
Product	Qurious — web application and installable Android APK
Compared against	IBM Quantum Learning + Composer, Qiskit Studio, Quirk, Quantum Odyssey, PennyLane Codebook, Brilliant, and eight further tools surveyed
Basis	Independent competitive research (Sept 2026) plus measured results from the working build

1. The Short Version

The quantum-education space is not empty, and any submission claiming otherwise will be corrected in the first question. It divides into five clusters, and each is genuinely strong on one axis.

Cluster	What it optimises for	Leading examples
Vendor curriculum + hardware	Rigour, real hardware access, credibility	IBM Quantum Learning & Composer, Microsoft Quantum Katas, AWS Braket
Browser circuit simulators	Speed, zero friction, quick tinkering	Quirk, Q.js, Quantastica Programming Studio
AI-assisted circuit builders	Natural language to circuit	Qiskit Studio (AI4quantum) — the closest analogue to this project
Gamified / puzzle learning	Motivation and intuition without heavy maths	Quantum Odyssey (Steam + Essentials mobile), Hello Quantum
SDK-attached coding courses	Writing real, runnable framework code	PennyLane Codebook, Brilliant, Qiskit tutorials

What does not exist anywhere is a single platform where all of these share one verified, gradable circuit representation. Each product solves one slice well. None combine a synced canvas-and-code builder, multi-SDK execution, an AI tutor whose output is checked against a simulator before the learner sees it, state-based grading, a learner model, and an offline mobile app.

> **The claim, stated precisely.** Qurious is not the first drag-and-drop simulator, the first AI-assisted builder, or the first mobile quantum app. It is the first to put them behind one circuit representation where every claim is verifiable — and, as of this build, the only one that also shows a learner what their circuit does on a noisy real machine.

2. Full Feature Comparison

Rows are the capabilities the problem statement asks for. "Partial" means a weak or indirect version exists. Every row was checked against the product’s live documentation.

2.1 Capabilities the problem statement requires

Capability	IBM Learning + Composer	Qiskit Studio	Quirk	Quantum Odyssey	PennyLane Codebook	Qurious
Drag-and-drop circuit canvas	Yes	Yes	Yes	Partial	No	Yes
Code editor two-way synced to the canvas	No	No	No	No	No	Yes
Multi-SDK execution (Qiskit / PennyLane / Cirq)	No	No	Partial	No	No	Yes
Structured lesson curriculum	Yes	No	No	Partial	Yes	Yes
AI tutor or explanations	No	Partial	No	No	No	Yes
AI output verified against a simulator first	No	No	No	No	No	Yes
Auto-grading by comparing produced vs target state	No	No	No	No	Partial	Yes
Adaptive / personalised learning path	No	No	No	Partial	No	Yes
Instructor dashboard	No	No	No	No	No	Yes
Runs offline, on-device, no server	No	No	Yes	Partial	No	Yes
Installable Android APK	No	No	No	Yes	No	Yes
Free and open source	Partial	Yes	Yes	No	Yes	Yes
Real quantum hardware access	Yes	No	No	Partial	No	No

2.2 Capabilities Qurious added that nothing surveyed has

These rows are not in the original research matrix because no competing product scored on them. They are what this build added on top of meeting the brief.

Capability	IBM Learning + Composer	Qiskit Studio	Quirk	Quantum Odyssey	PennyLane Codebook	Qurious
Noise simulation — see your circuit on a real, error-prone machine	No	No	No	No	No	Yes
Animated explanations computed live by the simulator	No	No	No	Partial	No	Yes
Offline knowledge base answering questions with no internet	No	No	No	No	No	Yes
Spoken tutor with a 3D avatar and wake word	No	No	No	No	No	Yes
Indian-language voice support	No	No	No	No	No	Yes
Prediction questions graded by live simulation, no answer key	No	No	No	No	No	Yes
Simulator numerically cross-verified against Qiskit	n/a	No	No	No	No	Yes
Lessons and challenges authorable as data, not code	No	No	No	No	No	Yes

Score across both tables: Qurious answers 20 of 21 capabilities. The single exception is real quantum hardware access, and Section 5 explains why that is a deliberate choice rather than a gap.

3. Product by Product

IBM Quantum Learning + Quantum Composer

The vendor benchmark — the best free curriculum in the field, and the tool this project must answer to.

	They do this well	Where they stop
	Free guided courses written by IBM's own education team, from fundamentals to utility-scale algorithms. Drag-and-drop Composer with live OpenQASM and Qiskit generation, q-sphere and histogram views. Direct execution on real superconducting hardware. Unmatched credibility.	Content is fundamentally read-and-watch. Exercises are multiple-choice or self-checked, never graded by executing the learner's own circuit. No two-way sync between code and canvas — they are separate surfaces. No learner model, so every student walks an identical path. Desktop-oriented, with no offline mobile app. Locked to the IBM and Qiskit stack.

> **Qurious answer.** We are not trying to out-write IBM's textbook. We close the gap between reading it and being able to build with it: a live circuit inside the lesson rather than a notebook beside it, grading on the actual quantum state, a path that adapts, and an offline app in the learner's pocket.

Qiskit Studio (AI4quantum)

The closest existing analogue — a visual, AI-assisted, web-based circuit builder with an educational purpose.

	They do this well	Where they stop
	Node-based drag-and-drop canvas for building Qiskit circuits. Deep AI integration for circuit generation and code assistance using agent tooling. A library of Qiskit patterns and demos to explore.	Explicitly a proof of concept. No structured curriculum, no assessment, no grading, no progress tracking, no instructor view, no learner model. Critically, no verification loop — AI output is shown without being checked, so it can be confidently wrong. Qiskit-only. No offline or mobile support.

> **Qurious answer.** This is the product we are measured against, and the difference is the verification loop. Qurious executes every AI-proposed fix on the simulator and discards it if the result does not match the claim. On a Bell-state repair it evaluated 36 candidate circuits and rejected 35 — those 35 are wrong answers a learner never saw.

Quirk

The fastest way to see a quantum state change — a toy, not a course.

	They do this well	Where they stop
	Instant, zero-friction, entirely client-side simulation with live visual feedback. Open source and free, extremely fast to experiment in. Cited as a reference tool inside academic quantum-education literature.	No lessons, no curriculum, no explanation of why a gate does what it does. No grading, no progress tracking, no accounts. No code editor or SDK export workflow. No AI assistance of any kind. No mobile app.

> **Qurious answer.** Qurious keeps Quirk's instant on-device feel — our simulator runs in the same tab with no server — and wraps a curriculum, grading, a tutor and a mobile app around it.

Quantum Odyssey (and Essentials mobile)

A gamified, puzzle-based approach — the strongest existing answer to "make it fun", weak on structured teaching.

	They do this well	Where they stop
	Over 500 open-ended optimisation puzzles needing no maths or coding background. Visualises the underlying linear algebra directly. Solutions are executable on real IBM hardware.	Game-first: no formal lesson structure, textbook content or instructor tooling. No AI tutor — it is a puzzle engine. No code editor or multi-SDK workflow. Progress is puzzle

	They do this well	Where they stop
	A genuine free mobile app already exists. Community layer where players optimise each other's solutions.	completion, not competency tracking. The full product is commercial; the mobile version is a stripped-down entry point.

> **Qurious answer.** Qurious takes the motivation lesson seriously — animated explanations, a speaking avatar, celebration on a correct answer — but keeps competency tracking rather than puzzle counts, and stays free and open source throughout.

PennyLane Codebook and Brilliant

SDK-real coding courses — you learn by writing actual framework code, chapter by chapter.

	They do this well	Where they stop
	Codebook is genuinely hands-on, executing real PennyLane in the browser, free, with auto-checked exercises. Brilliant offers strong visual pedagogy and guided problem sets with a browser-simulated quantum computer.	Codebook is PennyLane-only and code-first, with no visual canvas. Brilliant is a paid subscription and not open source. Neither has an AI tutor grounded in circuit verification. Neither works offline or on mobile. Grading is answer-checking on fixed exercises, not comparison of a learner's free-form circuit output against a target state.

> **Qurious answer.** Qurious exports to all four ecosystems from one circuit and lets the learner edit either the canvas or the code, because both are views of the same structure.

4. The Eight Things Nothing Else Does

Four were identified in the original research. Four more were built afterwards. Each is stated so a judge can check it.

4.1 Grounded AI tutoring

Every AI explanation or fix is executed on the simulator and thrown away if it does not produce the claimed result. IBM Composer and PennyLane have no AI tutor at all; Qiskit Studio generates but never verifies. Nothing surveyed checks AI output against ground truth before display.

Measured: repairing a half-built Bell state, the tutor evaluated 36 candidate circuits and discarded 35. The rejection count is shown to the learner deliberately.

4.2 One circuit representation driving everything

The canvas, the code editor, four SDK back ends and the auto-grader all read and write the same JSON structure. That is what makes true two-way canvas/code sync possible — IBM keeps the two surfaces separate, and Qiskit Studio is Qiskit-only.

4.3 Grading on physics, not multiple choice

Challenges are marked by comparing the learner's actual produced quantum state against a target, ignoring global phase. A different but correct solution passes, which a gate-by-gate comparison would wrongly fail. IBM uses self-checked exercises; Quantum Odyssey uses puzzle completion.

4.4 Offline, on-device mobile quantum sandbox

The whole learning loop — simulator, lessons, challenges, quizzes and the tutor's knowledge — runs with no connection at all. Quantum Odyssey: Essentials is the only other genuine mobile product found, and it is a puzzle game, not a curriculum-and-SDK learning tool.

4.5 NEW — Noise Lab: your circuit on a real machine

Every tool surveyed shows ideal quantum mechanics: perfect gates, perfect measurement, coherence that lasts forever. A learner who moves to real hardware meets decoherence with no intuition for it at all.

Qurious runs the same circuit the way a real machine would, using Monte Carlo quantum trajectories, and puts the ideal and noisy distributions side by side with four presets — ideal, trapped ion, today's superconducting, and early hardware.

> **What a learner sees.** A Bell state on today's superconducting preset: ideally $|01\rangle$ and $|10\rangle$ are impossible. With noise they appear 63 times in 1,024 shots, the match with ideal falls to 96.8%, and 10% of shots are hit by an error. Add two-qubit gates and watch it collapse further. This is the fastest possible explanation of why circuit depth is the constraint and why error correction is the field's central problem.

4.6 NEW — Animated explanations computed by the simulator

Ten animated walk-throughs across the lessons, 35 beats in total. Every frame is computed from the gates applied so far rather than drawn as a stored keyframe, so an animation can never disagree with the physics, and editing the gate list edits the animation. In the entanglement explainer the $|01\rangle$ bar visibly collapses to zero as $|11\rangle$ grows in its place.

4.7 NEW — A tutor that answers with no internet, and says which brain answered

105 quantum topics answerable entirely offline, spanning core concepts, gates, algorithms, mathematics, real hardware, error correction, applications, and careers. When a question falls outside that base and a language model is connected, the answer is labelled as coming from the model rather than the verified knowledge base. When neither can answer, it says so instead of guessing.

4.8 NEW — Spoken, multilingual, and physically present

A 3D avatar tutor that floats over the app, reacts to what the learner does, speaks its answers aloud, listens for questions, and responds to a "hey tutor" wake word — in nine languages including Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi and Bengali. Nothing in the survey has a voice interface at all, and nothing addresses Indian languages.

5. The One Row Where We Lose

Real quantum hardware access. IBM runs your circuit on an actual superconducting machine; Quantum Odyssey exports puzzle solutions to one. Qurious does not, and this is a deliberate trade rather than an unfinished feature.

The reasoning is about what a beginner actually needs. A learner building their first hundred circuits needs thousands of instant, free, forgiving runs. Real machines are queued, remote, rationed and noisy — a handful of jobs a day, each taking minutes to return. Optimising for hardware access would mean an account, a network connection and a wait, which is precisely the friction that keeps students out.

Simulating on the learner's own device removes the queue, the account, the cost, the latency and the internet. That is the decision that makes the offline Android app possible at all, and it is the reason the whole platform works on a phone with no connection.

> **The strongest form of this answer.** We did not skip hardware because it was hard. We skipped it because the Noise Lab teaches what hardware would teach — that real machines are imperfect — without the queue. A learner who has watched their Bell state break under a realistic error model understands more about hardware than one who queued a job and got a number back. Connecting real backends is a documented next step, not a missing pillar.

6. Evidence for Every Claim

Nothing in this document is asserted without a way to check it. These figures come from the working build.

Claim	How it was verified
The simulator is numerically correct	Cross-checked against Qiskit Aer on 22 circuits — Bell, GHZ, Toffoli, Fredkin, Grover, rotations, mixed 3-qubit and 4-qubit chains. Worst disagreement 6.661e-16, i.e. floating-point noise. Reproduce with: python server/crosscheck.py
The AI never shows an unverified fix	146 automated tests, including one asserting that a proposed repair actually produces the target state when run. Live: 36 candidates evaluated, 35 rejected.
Grading accepts different but correct answers	A test builds the same Bell state with the control on the other qubit and asserts it passes.
The noise model is physically sensible	17 tests: the ideal preset reproduces the exact ideal distribution; readout error flips bits at the stated rate; fidelity falls monotonically as gate error rises; error accumulates with depth; presets rank trapped ion above superconducting above early hardware.
The tutor does not answer questions nobody asked	A regression test asserts that typing "hi" matches no topic — it previously matched "Which qubit is which" because "hi" is a substring of "which".
It genuinely runs offline	Simulator, lessons, challenges, quizzes and the 105-topic knowledge base need no network. Only the optional language model and speech recognition do.
The APK is real	Built and installable, package in.sih.qurious, minSdk 22, containing the model, animation clips and the full offline curriculum.

7. The Hardest Question, and the Answer

> **"IBM already gives this away for free — why does your platform need to exist?"** IBM's courses are excellent, free and authoritative, and we studied them closely. But they remain something you read and watch: code that runs in a notebook beside the lesson rather than a live circuit inside it, exercises that are self-checked rather than state-verified, one identical path for every learner, and a desktop-first tool with no offline mobile app. Qurious is not trying to out-write IBM's textbook. It is closing the gap between reading that textbook and being able to build with it.

Say the second half out loud. Claiming to beat IBM on content invites a laugh. Claiming to close the practice loop around content like theirs is defensible, accurate, and exactly what the feature matrix shows.

8. Sources

Re-verify every link before presenting; product pages in this space change often.

- IBM Quantum Learning & Composer — quantum.cloud.ibm.com/learning
- Qiskit Studio (AI4quantum) — github.com/AI4quantum/qiskit-studio
- Quirk — algassert.com/quirk
- Quantum Odyssey — store.steampowered.com/app/2802710 and "Quantum Odyssey: Essentials" on Google Play
- PennyLane Codebook — pennylane.ai/codebook
- Brilliant Quantum Computing — brilliant.org/courses/quantum-computing
- Quantastica Quantum Programming Studio, Q.js, Microsoft Quantum Katas, Classiq, Quantum Flytrap Virtual Lab, Hello Quantum — surveyed, none change the picture

Academic grounding:

- IBM's quantum-verified rewards for Qiskit code generation — arXiv:2508.20907 (published support for our verification loop)
- Interactive visual circuit simulator courses — arXiv:2404.10328
- Quantum Composer, visualisation for education — arXiv:2006.07263
- Qiskit HumanEval benchmark — arXiv:2406.14712

Compiled from the team's independent competitive research (September 2026) and measured against the working Qurious build. Feature claims about competing products reflect their documentation at the time of writing and should be re-checked before submission.